ACPolyp: Active-Contour-Guided Feature Extraction for Endoscopic Finding and Polyp Classification

A leakage-audited, shortcut-aware study on Kvasir v2 — full technical report

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Abstract

Research prototype. Not a medical device. Kvasir is licensed for research and education only.

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Executive Summary

Question. Does an *unsupervised* active-contour prior improve deep classification of endoscopic findings on Kvasir v2, and through which mechanism?

Answer. No measurable improvement. The full contour-guided model scores macro-F1 0.9277 against 0.9233 for plain ConvNeXt-T ($\Delta = +0.0045$, 95 % CI -0.0004 to $+0.0091$, Holm-adjusted $p = 0.296$); controls with contours from the *wrong* image (+0.0043) or an image-independent ellipse (+0.0035) are indistinguishable. The contour itself is a weak localiser (Dice 0.406 against 0.375 for a central ellipse).

What the project delivers.

- A verified **data audit** of Kvasir v2: 31 duplicate frames (one filed under two classes), 6.9 % suspected label errors, resolution alone predicting upper-GI vs colon at 96.8 %, a scene-grouped leakage-aware split.
- A **benchmark** of five fine-tuned backbones (macro-F1 0.8708–0.9344; soft-voting ensemble 0.9385; polyp-vs-rest AUROC above 0.99), classical baselines and frozen-embedding probes under one protocol.
- A **controlled ablation** with paired bootstrap, McNemar, DeLong and Holm correction, learning curves, robustness curves and a learning-rate sweep.
- **Calibrated, conformal outputs**: ECE 0.046→0.033; group-aware conformal sets with 95.2 % coverage at a 95 % target (validation-calibrated sets under-covered at 92.9 % — an instructive failure).
- **Error analysis:** 73 % of errors fall in two confusable pairs and 53 % involve flagged labels, so label ambiguity bounds attainable accuracy.
- A tested, reproducible **software package** (48 tests, one-command pipeline, versioned model bundle), a local **web application**, an IEEE-format **paper**, this report and a slide deck.

Honest limits. Single-centre data without patient IDs; three seeds for deep models; one training recipe for all variants; no external validation; not clinical. Commercial use of Kvasir-trained models requires written permission from the dataset owner (Simula); this report states that plainly because it affects any plan to monetise the work.

Chapter 1

Problem Formulation, Plan and Governance

1.1 Domain and why a learned model

Colorectal polyps are mucosal outgrowths, some of which become cancer; detecting and characterising them during colonoscopy is a recognised quality indicator, and detection rates differ between endoscopists. A rule such as "red convex blob" fails because polyps are flat, pedunculated or dyed, mucosa varies in colour and folds, specular highlights and fluid create convex red regions everywhere. A hand-written colour rule (nearest Lab-colour centroid) reaches macro-F1 0.476 and polyp AUROC 0.837 (chance: 0.131 and 0.5), while fine-tuned networks exceed 0.9 and 0.99: a learnable statistical pattern exists and a deterministic rule is inadequate.

1.2 Learning problem

Supervised single-label classification into the 8 Kvasir classes, $f_\theta : \mathcal{X} \to \Delta^7$ with $\mathcal{X} = \mathbb{R}^{3\times H\times W}$. The clinical question "is there a polyp?" is read from the same posterior as $s(x) = P(y = \text{polyps} \mid x)$ with a validation-tuned threshold τ^*: a one-vs-rest task with a $1:7$ imbalance and, as hard negatives, the other colon classes. The dataset's "polyps" class is one of eight, so 8-class scores are never called polyp accuracy.

1.3 Objectives, research questions and hypotheses

Primary objective: quantify whether an unsupervised active-contour prior improves classification beyond a strong fine-tuned backbone, and attribute any effect to a mechanism. **Engineering constraints:** one 8 GB GPU, batch-1 latency below 25 ms, one-command reproducibility, a label-free and unit-tested contour stage.

RQ1 Does contour guidance change macro-F1 over plain ConvNeXt-T? $H_0 : \Delta = 0$ vs $H_1 : \Delta \neq 0$ (paired bootstrap, McNemar, DeLong, Holm).

RQ2 Is any gain due to the image-specific contour, to extra hand-crafted features, or to any mask? H_0: shuffled-contour and ellipse controls equal the full model.

RQ3 How much apparent accuracy is due to acquisition shortcuts and split leakage?

RQ4 Does guidance help more with little training data?

RQ5 Are probabilities calibrated, and do conformal sets attain nominal coverage?

A null result for RQ1 was declared acceptable in advance; the fallback was to report it and keep the audit, calibration and shortcut findings as the contribution.

1.4 Metric taxonomy

Primary: macro-F1 (8 classes) and AUROC of P(polyps). Diagnostic: accuracy, balanced accuracy, MCC, per-class precision/recall/F1, sensitivity/specificity/PPV at τ^*, NLL, Brier, ECE (15 bins). Uncertainty: conformal coverage and set size, risk–coverage. Efficiency: parameters, peak VRAM, training time, batch-1 latency after warm-up with CUDA synchronisation, checkpoint size. Contour: Dice and IoU against Kvasir-SEG masks. Cost-sensitive: expected cost per image with a miss costing five false alarms.

1.5 Feasibility audit and scope

Data: 8,000 labelled frames plus 1,000 masks, openly downloadable. **Hardware:** RTX 4060 Laptop (8 GB), i7-13650HX, 23.7 GB RAM, 195 GB free disk. **Cost:** ConvNeXt-T at 224 px, batch 32, bf16: about 25–30 s per epoch and 2.5 GB peak VRAM when the machine is otherwise idle (several times slower when other jobs share it, as happened here). **In scope:** Kvasir v2, Kvasir-SEG as an auxiliary contour benchmark, timm backbones, classical baselines, calibration, conformal prediction, explainability, a local demo. **Out of scope:** detection or segmentation products, video, clinical validation, regulatory claims, and HyperKvasir (its frames overlap Kvasir, so it is not an independent test).

1.6 Risk register

ID	Risk	Mitigation	Outcome
R1	Download truncation	byte count, SHA-256, zip test, resume	occurred (771 MB of 2.49 GB), caught
R2	Duplicates, label contradiction	hash and embedding de-duplication, data contract	occurred, fixed
R3	Null result for contour guidance	controls, pre-declared fallback	**occurred**; reported
R4	Shortcut learning	canonicalisation, naive-vs-canonical run	shortcuts exist; no accuracy effect
R5	Split leakage	scene groups, random-split control	mitigated; AUROC inflated by random split
R6	OOM / commit memory	bf16, memory-mapped data, GPU augmentation	commit exhaustion hit once, fixed by removing workers
R7	Silent config coercion	parser and unit test	occurred in test, fixed
R8	Statistics bugs	tests against scikit-learn and hand cases	occurred (DeLong), fixed
R9	Over-claiming	numbers generated from logs, limitations section	ongoing
R10	Commercial misuse of research-only data	licence audit, no redistribution	documented

1.7 Licensing

Kvasir and Kvasir-SEG are restricted to *research and educational use*; commercial use is forbidden without prior written permission from Simula. The datasets are not redistributed; a few frames appear as attributed illustrative figure panels, as is usual in research papers. The code is MIT-licensed; any model trained on Kvasir and any product built on it inherit the dataset's non-commercial terms. Pretrained weights are Apache-2.0 (ConvNeXt-T, ResNet-50, EfficientNetV2-S, DINOv2) or MIT (Swin-T), verified from model-hub metadata. A portfolio, paper, teaching resource and open-source code are within the terms; selling or deploying the model commercially is not, without permission.

Chapter 2

Data Acquisition, Audit and Curation

2.1 Provenance and integrity

Kvasir v2 (8,000 JPEG frames, 8 classes × 1,000) and Kvasir-SEG (1,000 polyp frames with expert masks) were downloaded from the owners' server (`datasets.simula.no`) on 2 November 2026. The Kvasir v2 archive is 2,489,312,085 bytes, equal to the server's reported `Content-Length`; it passes Python's `zipfile.testzip()` and its SHA-256 is recorded in `data/raw/MANIFEST.json`. Extraction goes into an empty directory and rejects absolute paths and `..` components, because archives are untrusted input.

An incident worth recording. The first transfer silently stopped at 771 MB: the server closed the connection and the shell pipeline hid `curl`'s non-zero exit status. The size check caught it, and the final downloader (`data/download.py`) resumes with HTTP Range requests until the byte count matches exactly.

2.2 Defects discovered

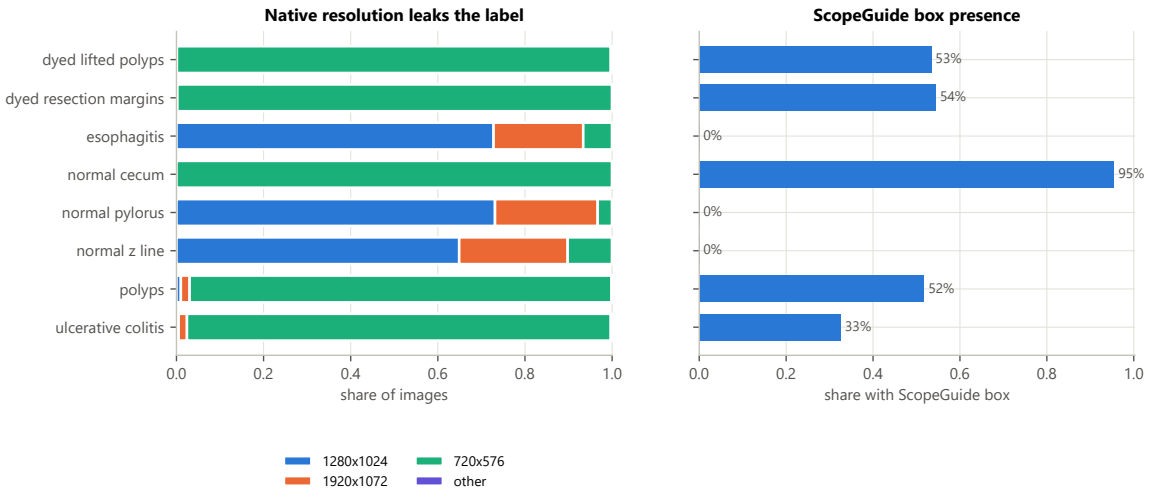


Figure 2.1: Acquisition artefacts by class: native-resolution mix (left) and presence of the ScopeGuide box (right).

Table 2.1: Data defects found in Kvasir v2 and the action taken.

#	Finding	Action
1	One file is byte-identical in two different classes (`dyed-lifted-polyps` and `dyed-resection-margins`)	both copies dropped
2	11 more byte-identical pairs; embedding duplicates (cosine ≥ 0.98) bring the total to 30 pairs	first kept; 31 rows removed, 7,969 remain
3	Native resolution leaks the label: colon classes are 97–100 % 720×576, upper-GI classes 90–97 % larger formats	FOV crop and square resize; leak measured at 96.8 % (majority 62.4 %)
4	ScopeGuide picture-in-picture box in 95 % of cecum frames, about 52 % of polyp frames, none of the upper-GI frames	canonical rectangle masked in *every* frame
5	First version of the box detector also hid blue dye (a lesion cue)	stricter colour, flatness and placement test; re-verified visually
6	Burned-in overlay text; no EXIF personal-data fields in any file (0/8,000)	reported as a limitation
7	No patient or video identifiers	scene groups from embedding similarity ($\geq$ 0.92)
8	Perceptual hashing fails: the shared octagonal border dominates low-frequency bits (at Hamming $\leq$ 2, 63 of 134 pairs were cross-class)	pHash only for exact matches
9	Confident learning flags 550 labels (6.9 %), concentrated in esophagitis, z-line and the two dyed classes	flagged, not removed

2.3 Canonicalisation

The endoscopic field of view is the largest bright blob after morphological opening (which strips thin text strokes); its convex hull gives the FOV mask and the crop box. A rectangle covering the ScopeGuide location (measured on 419 detected boxes) is blacked out in every frame, and the *valid* mask excludes FOV exterior and this rectangle. Figure 2.3 shows the stages on three frames.

2.4 De-duplication, scene groups and splits

Frozen DINOv2-S/14 embeddings give cosine similarity. Inspection of sampled pairs showed that similarity ≥ 0.95 is the same frame, while 0.88–0.95 is the same lesion or lumen from a slightly different pose (consecutive frames of one procedure). Single-linkage at 0.90 chains into a 546-image blob, so the group threshold is 0.92, giving 7,812 groups (largest 7). A stratified group split yields 5,578 / 797 / 1,594 train / val / test images with zero group or hash overlap. A naive stratified random split with the same sizes would put 41 groups on both sides of train/test; 2.7 % of its test images have a twin in the training set at similarity ≥ 0.92. Near-duplicate leakage is therefore modest, and we report that size honestly.

2.5 Distributions and collinearity

Mean luma is close to normal (Shapiro–Wilk $p = 0.16$), Laplacian variance is right-skewed (skew 1.36) and used on a log scale. The 98 contour-derived features are heavily collinear (median

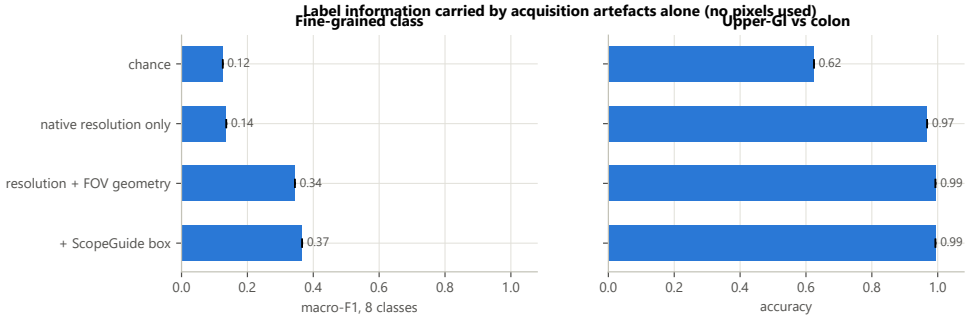


Figure 2.2: Label information available from acquisition artefacts alone, using no pixel values. A gradient-boosted classifier on native resolution separates upper-GI from colon frames with 96.8 % accuracy, and with FOV geometry 99.3 %.

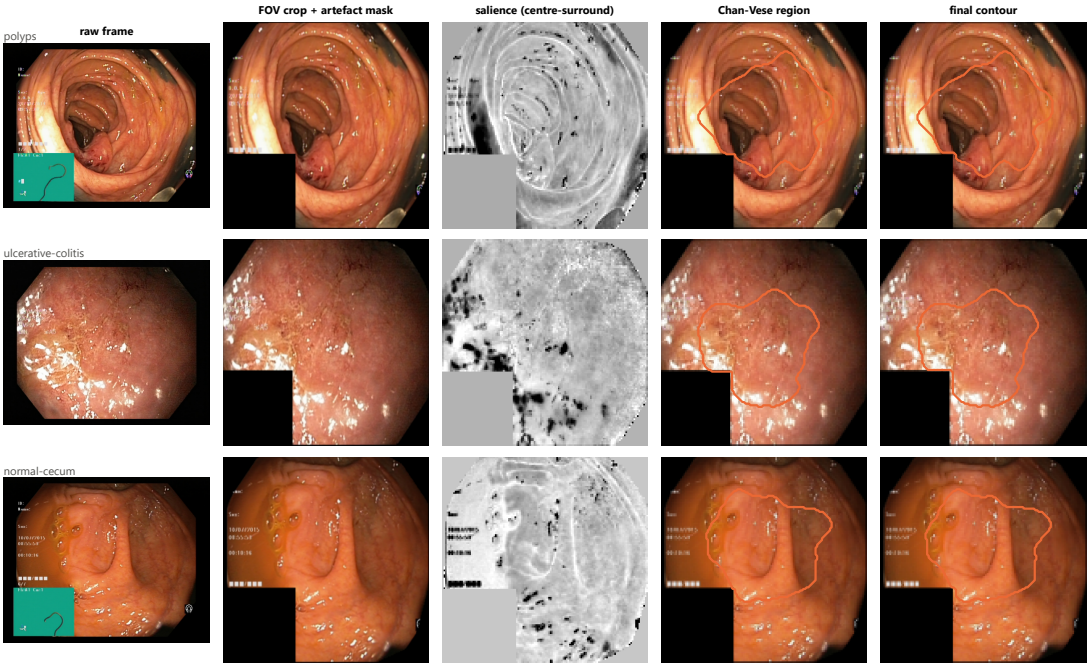


Figure 2.3: Pipeline illustration on a polyp, a colitis and a cecum frame: raw frame, FOV crop with artefact mask, centre–surround salience map, Chan–Vese region and final contour.

VIF 220, 79 features with VIF above 10), so regularised linear models and trees are used and individual coefficients are never interpreted.

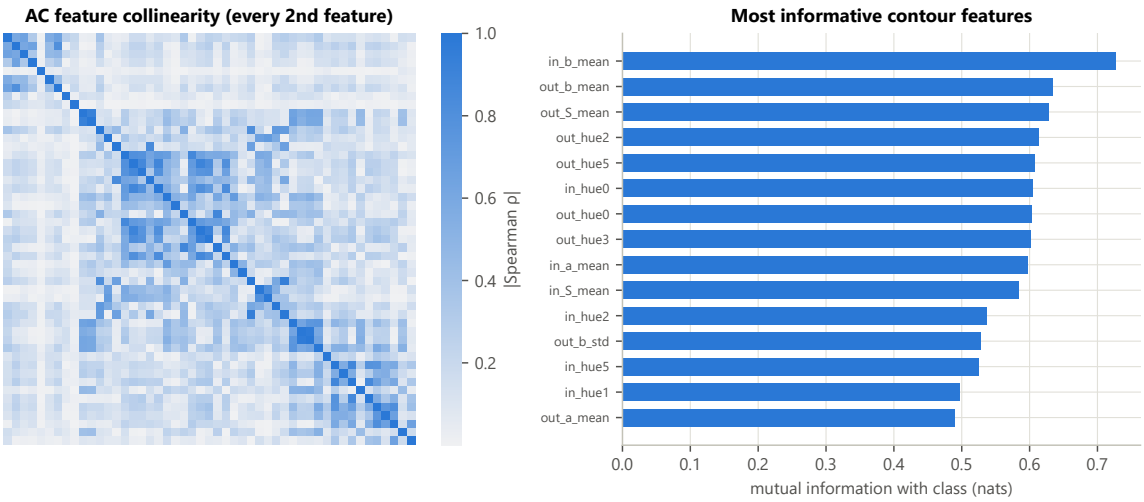


Figure 2.4: Left: collinearity of the contour features. Right: mutual information of the most informative features with the class.

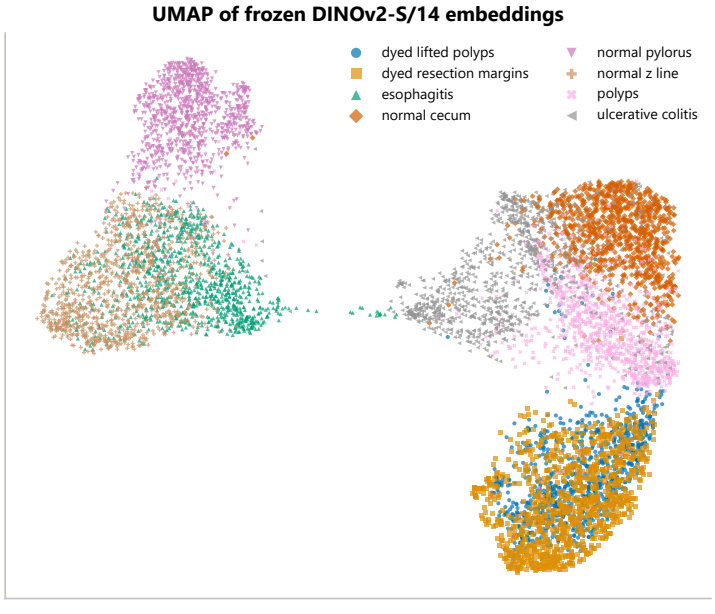


Figure 2.5: UMAP of frozen DINOv2 embeddings coloured by class (distinct marker shapes as a second channel).

Chapter 3

The Study: Related Work, Method, Experiments and Results

3.1 Related Work

Gastrointestinal datasets. Kvasir [19] introduced 8 expert-labelled classes; Kvasir-SEG [11] added 1,000 polyp frames with expert masks, and HyperKvasir [3] scaled the collection to 110,079 images and 374 videos from the same hospital. For video-derived collections, grouping frames by video is already recommended to avoid inflated test scores from near-identical neighbouring frames; Kvasir v2 provides no such identifiers, which is why we construct scene groups from embedding similarity (Section 3.2.1). Reported accuracies on Kvasir vary widely with the class subset, split and preprocessing, so we avoid claiming superiority over published numbers and instead compare all methods under one protocol.

Active contours. Snakes minimise an energy balancing smoothness and edge attraction [12]; the Chan–Vese model replaces edges by a region-based Mumford–Shah fitting term and handles weak boundaries [4]. Márquez-Neila et al. replaced PDE solvers by morphological operators on a binary level set, giving simple, fast and stable implementations [16]. A recent comparison on colonoscopy frames found Chan–Vese to fit polyp outlines better than the geodesic model when initialised inside the polyp, and that artefact removal and filtering matter [10]. These works study *segmentation*; we study the contour as a spatial prior for *classification* and measure it against a trivial centre prior.

Backbones. We benchmark ResNet-50 with an improved training recipe [8, 23], EfficientNetV2 [21], ConvNeXt [15], Swin [14] and a DINOv2 ViT [18], all loaded from `timm` [22].

Shortcuts and label noise. Deep networks often exploit decision rules that work on a benchmark but not under distribution shift [6]. Confident learning ranks suspicious labels from out-of-sample predicted probabilities [17]; we use it to quantify, not to remove, label noise.

Calibration and conformal prediction. Modern networks are often over-confident; temperature scaling is a simple fix [7]. Split-conformal prediction yields sets with finite-sample marginal coverage [2]; we use both on a clinically motivated question: when should the system defer to the endoscopist?

3.2 Method

3.2.1 Data curation and splits

Canonicalisation. Frames are cropped to the endoscopic field of view (FOV; largest bright blob after morphological opening, convex hull), resized to 256×256, and a canonical rectangle covering the ScopeGuide box location (measured on 419 flagged frames) is blacked out in *every* frame, regardless of native resolution, so that neither the box nor the resolution is available to the network. Pixels outside the FOV or inside the rectangle form the invalid mask.

De-duplication and scene groups. Frozen DINOv2-S/14 embeddings define cosine similarity. Pairs with similarity ≥ 0.98 (or identical bytes) are duplicates: within a class the first is kept; cross-class pairs are dropped. Single-linkage components at similarity ≥ 0.92 form *scene groups* (0.90 already chains into a 546-image component). A stratified group split gives train / val / test of 5,578 / 797 / 1,594 images with zero group or hash overlap. A stratified random split with the same sizes serves as a leakage control.

3.2.2 Active-contour stage

On the salience image S (centre–surround of the first Lab principal component, 128 px, invalid pixels set to the valid mean) we run morphological Chan–Vese [4, 16] from a central disc; the region touching the FOV border least is the lesion candidate. A morphological geodesic active contour can then snap it to edges. Hyperparameters (channel, centre–surround scale, initial radius, iterations, smoothing, region weight, geodesic settings; 80 TPE trials [1]) were optimised for Dice against Kvasir-SEG masks of frames matched to the *training* split, selected on validation-matched frames and reported on test-matched frames. No classification label is used. From the final mask we compute 136 features in four families: *global* colour/texture over the valid FOV; *shape* (area, circularity, solidity, eccentricity, Hu moments, curvature statistics, border contact, centroid offset); *boundary* (edge strength and inside/outside ring contrast); *region* (Lab/HSV statistics, LBP and masked GLCM inside and outside).

3.2.3 ACGNet

Let $F \in \mathbb{R}^{C\times h\times w}$ be the last feature map of the backbone, V the valid mask and M the contour mask, both area-downsampled to $h\times w$. With layer normalisation LN,

$$g = \mathrm{LN}\Big(\frac{\sum VF}{\sum V}\Big), \quad f_{in} = \mathrm{LN}\Big(\frac{\sum MF}{\sum M+\epsilon}\Big), \tag{3.1}$$

$$f_{out} = \mathrm{LN}\Big(\frac{\sum (V-M)_+ F}{\sum (V-M)_+ +\epsilon}\Big), \tag{3.2}$$

$$e = \mathrm{GELU}\big(W_2\,\mathrm{GELU}(W_1 s)\big), \tag{3.3}$$

$$z = W\,\mathrm{Dropout}([g; f_{in}; f_{out}; e]), \tag{3.4}$$

where s is the standardised contour-feature vector (statistics fit on the training split only, 1/99-percentile winsorised). Regions with mass below 0.05 fall back to g. Ablation modes remove $[f_{in}; f_{out}]$ (*feature-only*), e (*pool-only*) or both (*plain*). **Controls:** *shuffled*—masks and features of a *different* image, preserving mask statistics but destroying alignment; *ellipse*—an image-independent central ellipse; *gfeat*—extra hand-crafted features without a contour. Masks follow the image through every geometric augmentation.

3.2.4 Training and post-processing

AdamW with weight decay 0.05 (not on biases or norms), backbone learning rate 0.25× the head rate ($2{\cdot}10^{-4}$), one warm-up epoch then cosine decay, label smoothing 0.1, gradient clipping 1.0, stochastic depth 0.1, bf16 autocast, 10 epochs, early stopping on validation macro-F1 (patience 4). Augmentation: random resized crop, flips, rotation by ±180°, colour jitter. After training, a temperature T minimises validation NLL [7]; the polyp threshold τ^* maximises Youden's J on validation; split-conformal sets use the least-ambiguous score $1 - p_y$ with $\alpha = 0.05$ [2].

3.3 Experimental Setup

Data. Kvasir v2 (8,000 frames, 8 classes × 1,000) and Kvasir-SEG (1,000 masked polyp frames), both used under their research-and-education terms. After de-duplication 7,969 frames remain. All test numbers are computed once per frozen model on the 1,594-image grouped test split.

Models. *Trivial:* random and class prior. *Heuristic:* nearest Lab-colour centroid. *Classical:* logistic regression, RBF-SVM, random forest, LightGBM [13] and XGBoost on the global and contour feature sets, hyperparameters tuned on validation with 25 Optuna trials; a linear probe on frozen DINOv2 embeddings. *Deep:* ResNet-50, EfficientNetV2-S, ConvNeXt-T, Swin-T, DINOv2 ViT-S/14, all fully fine-tuned from public checkpoints (Apache-2.0 / MIT), and the guided ACGNet variants on ConvNeXt-T and ResNet-50.

Protocol. Deep models: N=3 seeds (initialisation of new layers, data order, augmentation); classical models: N=5. We report mean ± SD. Differences between variants use (i) a paired bootstrap over test images with 2,000 resamples of the seed-averaged macro-F1, (ii) exact McNemar and DeLong [5, 20] tests on seed-ensembled predictions, (iii) a seed-level paired t-test, and Holm–Bonferroni correction [9] across the ablation family. With three seeds a Wilcoxon signed-rank test cannot fall below $p = 0.25$ (two-sided), so we do not rely on it.

Metrics. Macro-F1 (primary), accuracy, MCC, polyp-vs-rest AUROC/AUPRC, NLL, Brier, ECE (15 bins), conformal coverage, latency (batch 1, after warm-up, with CUDA synchronisation), peak VRAM.

Implementation and hardware. PyTorch, `timm 1.0.30`, scikit-image, LightGBM, Optuna; a single laptop RTX 4060 (8 GB) and i7-13650HX. Every run directory stores its configuration, git hash, command, environment, metrics and logits. The package contains 38 automated tests covering hand-computed metric cases, DeLong against scikit-learn, gradient flow, single-batch overfitting, batch-order invariance and the leakage-free split logic.

3.4 Results

3.4.1 Active-contour quality

On the 176 test-split polyp frames that could be matched to Kvasir-SEG masks, the tuned contour reaches a Dice of 0.406 (95 % bootstrap CI 0.373–0.440, IoU 0.281), against 0.375 for a central ellipse, 0.366 for the untuned defaults and 0.272 for the whole field of view. The paired gain over the ellipse is +0.031 Dice (CI +0.019 to +0.042, Wilcoxon $p = 2.5{\times}10^{-6}$); the contour is better on 62 % of the images. It is a *weak* localiser: it over-segments (median 2.4× the expert area) and fails on small lesions (Dice 0.21 for polyps under 10 % of the FOV, 0.45 for 10–25 %, 0.62 above

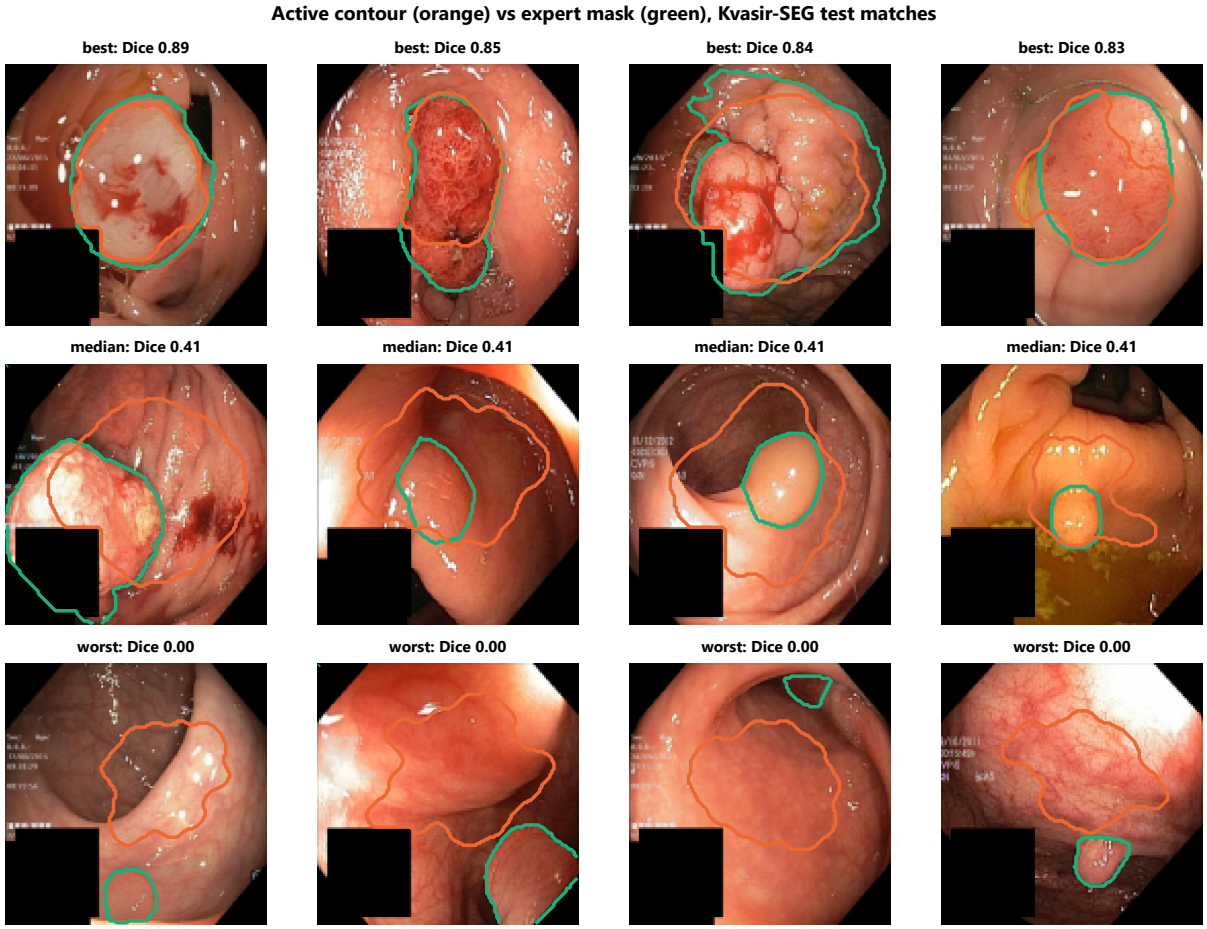


Figure 3.1: Contour (orange) vs expert mask (green) on Kvasir-SEG test matches: best, median and worst by Dice. Failures are lumen/fold regions and small lesions.

25 %; Fig. 3.1). The search itself switched off the geodesic refinement stage (0 iterations), i.e. the second stage did not help.

3.4.2 Classical baselines and feature controls

Chance is a macro-F1 of 0.131; the Lab-colour centroid heuristic reaches 0.476 (polyp AUROC 0.837). LightGBM on global colour and texture features reaches 0.751; adding the contour features raises it to 0.795. *However, the controls match or exceed the real contour:* replacing the contour by an image-independent central ellipse gives 0.800, and a contour borrowed from another image gives 0.784. What helps is therefore "colour and texture inside versus outside a central region", not the contour itself. A linear probe on frozen DINOv2 embeddings reaches 0.889, above every hand-crafted pipeline.

3.4.3 Deep backbones

Table 3.1 compares all models under one protocol. Fine-tuned ConvNeXt-T, EfficientNetV2-S, Swin-T and DINOv2 ViT-S/14 lie within about one to two points of each other (macro-F1 0.9233, 0.9257, 0.9327 and 0.9344), while the ResNet-50 recipe trails at 0.8708. Polyp-vs-rest AUROC is above 0.99 for all fine-tuned networks. A soft-voting ensemble of the five backbones reaches 0.9385.

Table 3.1: Held-out test performance (mean±SD over seeds; deep models N=3, classical N=5). Best per column in bold. ECE after temperature scaling.

Model	Macro-F1	Acc.	MCC	Polyp AUROC	ECE
ResNet-50	0.8708±0.0043	0.8712±0.0045	0.8530±0.0051	0.9916±0.0017	0.0300±0.0026
EfficientNetV2-S	0.9257±0.0025	0.9258±0.0025	0.9153±0.0029	**0.9957±0.0004**	0.0256±0.0035
ConvNeXt-T	0.9233±0.0006	0.9235±0.0006	0.9128±0.0007	0.9931±0.0029	0.0353±0.0027
Swin-T	0.9327±0.0045	0.9327±0.0045	0.9232±0.0051	0.9947±0.0014	0.0306±0.0050
DINOv2 ViT-S/14	**0.9344±0.0058**	**0.9343±0.0058**	**0.9250±0.0066**	0.9916±0.0015	0.0242±0.0035
ACG-ConvNeXt (full)	0.9277±0.0035	0.9279±0.0035	0.9178±0.0040	0.9900±0.0013	0.0325±0.0016
ACG-ResNet-50	0.8560±0.0103	0.8565±0.0100	0.8363±0.0114	0.9887±0.0023	0.0353±0.0049
LightGBM · global colour/texture	0.7507±0.0037	0.7513±0.0037	0.7160±0.0042	0.9565±0.0003	0.1285±0.0011
LightGBM · contour features only	0.8063±0.0029	0.8073±0.0029	0.7804±0.0033	0.9721±0.0005	0.1450±0.0036
LightGBM · global + contour	0.7954±0.0045	0.7964±0.0045	0.7678±0.0052	0.9716±0.0006	0.1310±0.0056
Logistic reg. · global + contour	0.7763	0.7773	0.7458	0.9712±0.0000	0.0308±0.0000
LightGBM · global + central-ellipse features (control)	0.8003±0.0028	0.8005±0.0028	0.7721±0.0032	0.9788±0.0002	0.1303±0.0027
LightGBM · global + shuffled-contour features (control)	0.7840±0.0013	0.7847±0.0014	0.7542±0.0016	0.9695±0.0004	0.1338±0.0008
RBF-SVM · global + contour	0.7900	0.7905	0.7609	0.9749±0.0000	**0.0219**
Random forest · global + contour	0.7658±0.0044	0.7666±0.0044	0.7338±0.0051	0.9624±0.0004	0.1079±0.0044
XGBoost · global + contour	0.7947±0.0045	0.7955±0.0044	0.7667±0.0051	0.9735±0.0011	0.1179±0.0054
Linear probe · frozen DINOv2	0.8889±0.0000	0.8890	0.8732	0.9951±0.0000	0.0354
Random guess	0.1311	0.1311	0.0070	0.4947	0.2116
Class prior (majority)	0.0279	0.1255	0.0000	0.4917	0.0000
Heuristic · nearest Lab-colour centroid	0.4762	0.4793	0.4066	0.8372	0.2669

Table 3.2: Ablation on ConvNeXt-T (N=3 seeds). Δ: seed-averaged macro-F1 difference to the plain baseline with 95 % paired-bootstrap interval over test images; p: Holm-adjusted bootstrap p-value.

Variant	Macro-F1	Acc.	Polyp AUROC Δ [95 % CI]	p_{Holm}
ConvNeXt-T (plain)	0.9233±0.0006	0.9235	0.9931 +0.0000 –	–
+ global hand-crafted features	0.9295±0.0035	0.9295	0.9927 +0.0062 [+0.0013, +0.0111]	0.072
+ contour pooling	0.9264±0.0037	0.9264	0.9934 +0.0031 [-0.0015, +0.0075]	0.380
+ contour features	0.9288±0.0003	0.9289	0.9932 +0.0055 [+0.0004, +0.0106]	0.180
ACG-ConvNeXt (full)	0.9277±0.0035	0.9279	0.9900 +0.0045 [-0.0004, +0.0091]	0.296
full, shuffled contours (control)	0.9275±0.0015	0.9276	0.9912 +0.0043 [-0.0010, +0.0093]	0.354
full, central ellipse (control)	0.9267±0.0067	0.9268	0.9908 +0.0035 [-0.0017, +0.0085]	0.380

3.4.4 Does contour guidance help? (RQ1, RQ2)

Table 3.2 and Fig. 3.2 give the ablation on ConvNeXt-T. Plain ConvNeXt-T scores 0.9233±0.0006. The full ACG model scores 0.9277±0.0035 ($\Delta = +0.0045$, 95 % CI −0.0004 to +0.0091, Holm-adjusted $p = 0.296$). **The difference is not statistically supported**, and neither are the shuffled-image control (+0.0043) and the ellipse control (+0.0035): all three overlap. Non-contour global features alone give +0.0062, the largest and only nominally significant bootstrap difference, which does not survive Holm correction (adjusted $p = 0.072$). Seed-level paired t-tests with N=3 are fragile (the plain model's seed SD is only 0.0006, which inflates standardised effect sizes) and are therefore reported in the repository but not relied on. Generality: ACG on ResNet-50 scores 0.8560 ± 0.0103 against 0.8708 for plain ResNet-50.

3.4.5 Shortcuts and leakage (RQ3)

Despite the strong metadata shortcuts (Section 3.2.1), a ConvNeXt-T trained on *naive* full frames (no crop, no artefact mask) reaches 0.9248 ± 0.0048, statistically equal to the canonical 0.9233: the shortcuts separate upper-GI from colon, which pixels already do, while the remaining errors are within-region pairs where neither shortcut carries information. Training and testing on a *random* split with the same sizes gives 0.9356 ± 0.0037 (different test images, so not paired), consistent with the modest 2.7 % near-duplicate overlap measured in the audit.

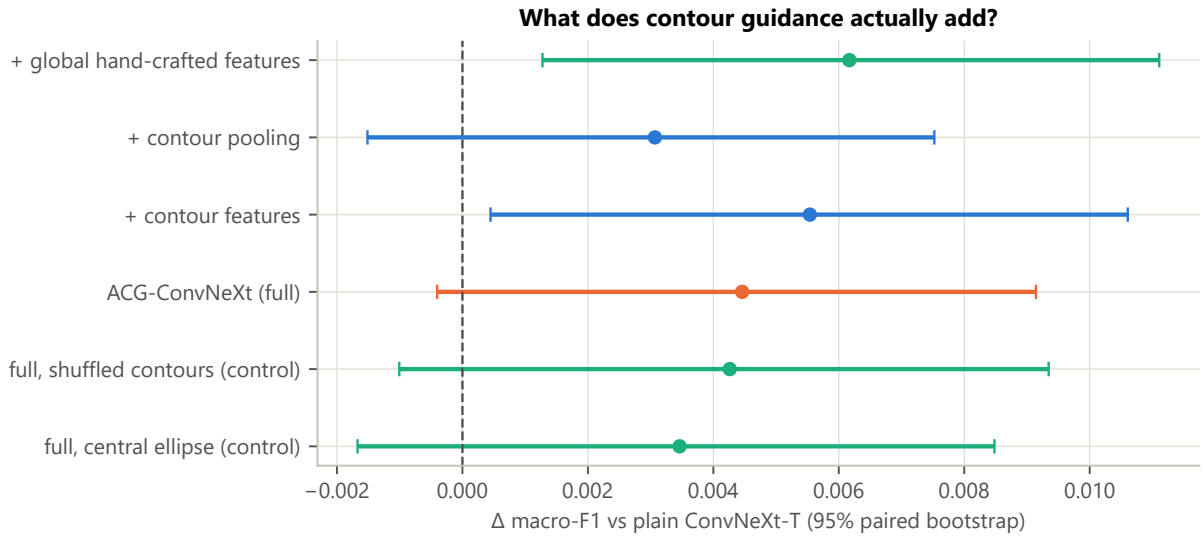


Figure 3.2: Difference in macro-F1 to plain ConvNeXt-T with 95 % paired-bootstrap intervals (unadjusted). The full model and both image-independent controls are indistinguishable.

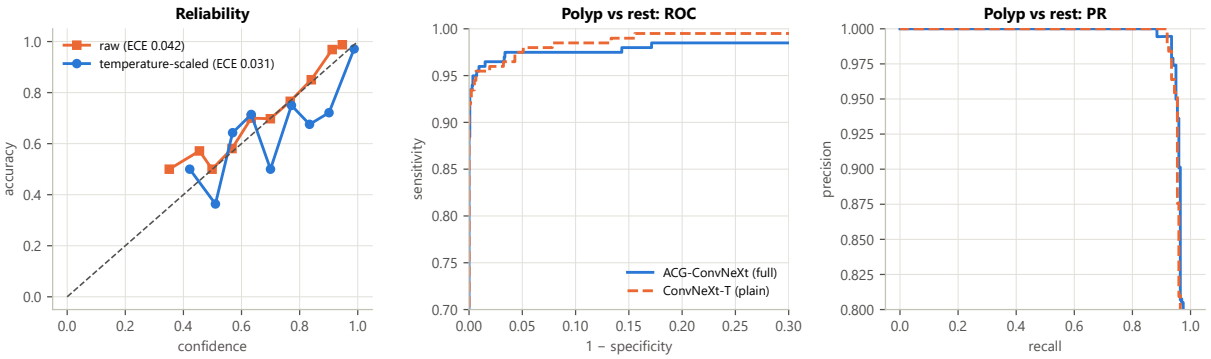


Figure 3.3: Reliability before/after temperature scaling, and polyp-vs-rest ROC and PR curves (seed 0).

3.4.6 Calibration and conformal prediction (RQ5)

Temperature scaling reduces ECE from 0.046 to 0.033. Validation-calibrated conformal sets reach only 92.9 % coverage at a nominal 95 %, because the validation split was reused for early stopping, temperature and threshold and is therefore optimistic. Calibrating on a random half of the test *scene groups* and evaluating on the other half (200 repeats) restores coverage: $95.2\,\% \pm 1.1\,\%$, mean set size 1.06, 94 % singleton sets (Fig. 3.3). For polyp-vs-rest, at a miss cost five times a false alarm, the 0.5 threshold has expected cost 0.040 per image against 0.037 for the validation-Youden threshold and 0.034 for the validation cost-optimal threshold: thresholds tuned on the optimistic validation split did not beat the default.

3.4.7 Error analysis

Of 117 test errors (7.3 %), 73 % fall into two confusable pairs (esophagitis↔z-line and dyed-lifted↔dyed-resection-margins; Fig. 3.4). 53 % of the errors involve images flagged by confident learning, against 5 % of correct predictions, so a large share of the residual error is label ambiguity rather than model failure. We inspected the 36 most confident errors (a visual judgement by a non-clinician, not an adjudication): 17 are esophagitis↔z-line, 7 are dyed-lifted↔resection-margin, 7

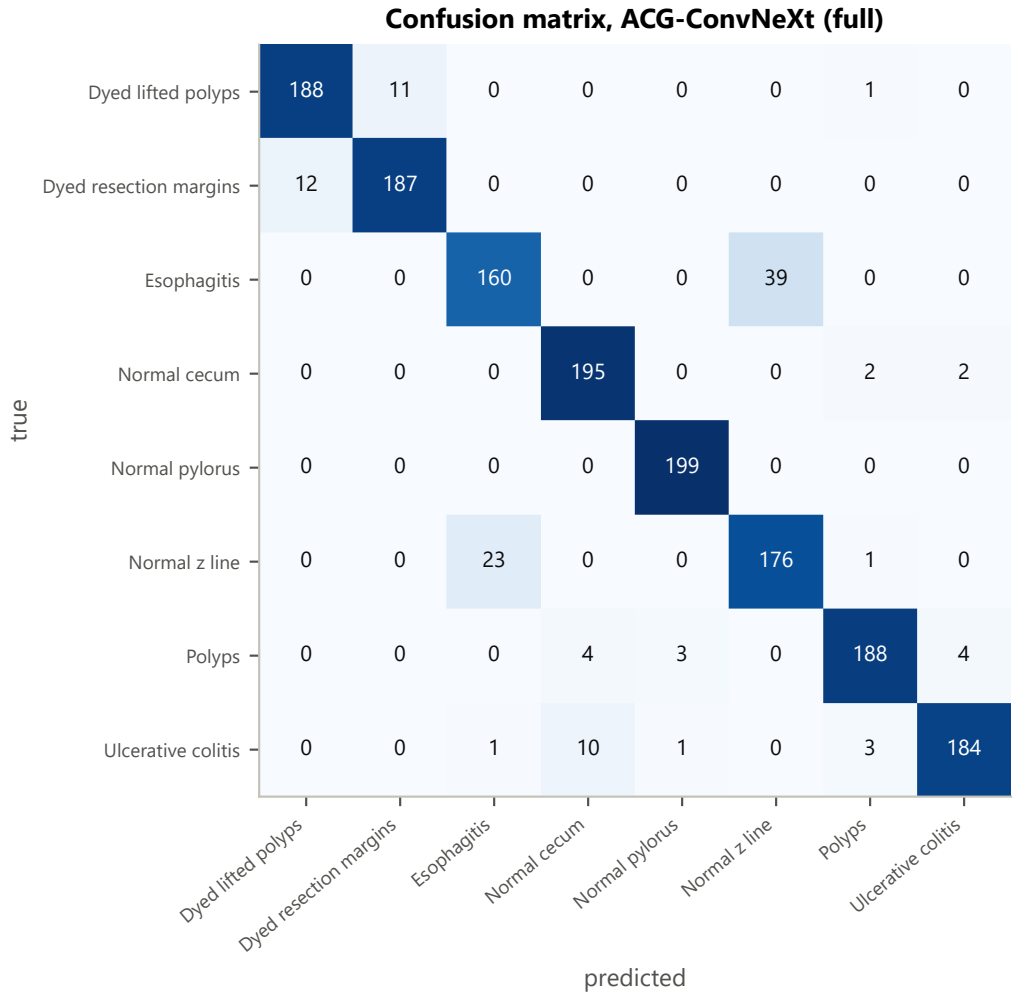


Figure 3.4: Confusion matrix of the seed-ensembled ACG-ConvNeXt on the test split.

are "polyp" frames with no distinct visible lesion (the clinically important group, only 2 of which are flagged), and 5 are other colitis/cecum mixes; 26 of the 36 are flagged. The most confident errors (Fig. 3.5) show craters with dye that are plausible for either dyed class, homogeneous mucosa labelled "polyp" with no visible lesion, and z-line frames with inflammation. Accuracy is 95.0 % on colon-scope (720×576) frames and 88.4 % on larger-format upper-GI frames; the worst class accuracy is 80.4 %.

3.4.8 Data efficiency, robustness and interpretability (RQ4)

Data efficiency (RQ4). Training on 10 %, 25 %, 50 % and 100 % of the training split gives macro-F1 0.8473, 0.8923, 0.9185, 0.9233 for the plain model and 0.8439, 0.8997, 0.9198, 0.9277 for ACG-ConvNeXt (Fig. 3.6). The curves overlap: guidance does not help more when data are scarce.

Learning-rate sensitivity. Deep hyper-parameters were not tuned per variant. For seed 0, the plain model scores 0.9168, 0.9227 and 0.9265 at half, default and double learning rate, and the guided model 0.9240, 0.9315 and 0.9385: changing the learning rate moves macro-F1 by as much as any ablation effect, and the default rate was evidently not optimal for the plain model. The guided model's seed-0 score at the doubled rate looked like a clear gain, but that rate was singled out after seeing test results, so we replicated it as an exploratory, post hoc check with

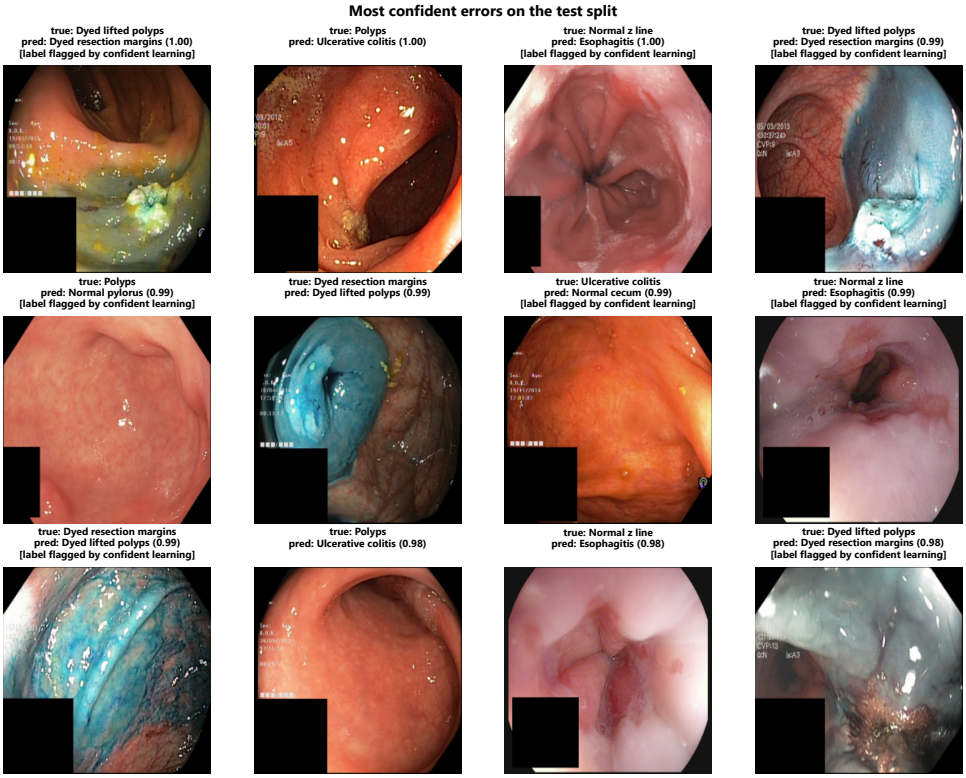


Figure 3.5: The twelve most confident errors, with confident-learning flags.

three seeds per variant: plain 0.9311 ± 0.0040, guided 0.9322 ± 0.0055 ($\Delta = +0.0011$, 95 % CI -0.0032 to $+0.0055$, unadjusted bootstrap $p = 0.63$). Both variants improve over the default rate and the gap between them is not distinguishable from zero, so the null conclusion holds at the better learning rate as well. A fully fair comparison would tune every variant on validation data, which we did not do.

Robustness. On 480 stratified test frames, JPEG compression at quality 50 lowers accuracy from 0.917 to 0.901 for the plain model, but only from 0.924 to 0.930 for the guided model (Fig. 3.7). At quality 10 the figures are 0.786 against 0.831, and under Gaussian blur with $\sigma = 3$ px 0.848 against 0.879. *The controls reproduce this advantage:* shuffled-image contours give 0.926 and the central ellipse 0.930 at quality 50, and global features alone give 0.922. Robustness therefore comes from supplying colour and texture statistics computed on the pixels, not from the contour's alignment with the lesion.

Interpretability. *Grad-CAM against expert masks.* Using the polyp-class Grad-CAM on the 176 Kvasir-SEG test-split polyps, the CAM maximum lies inside the expert mask for 84 % of frames for the plain model and 80 % for the guided model; the top-20 % CAM pixels overlap the mask with IoU 0.348 and 0.327. The CAM maximum falls inside the active contour for 65 % and 68 % of frames. With $n = 176$ these differences are within sampling noise: the guided network does not attend to lesions more accurately than the plain one. *TreeSHAP* on LightGBM with global and contour features attributes 58 % of the total importance to inside/outside colour and texture (region) features and 32 % to global features, but only 6 % to contour shape and 3 % to boundary features: the geometry of the contour carries little information. *Layer-wise linear probes* on ConvNeXt-T stages give macro-F1 0.819 (pretrained) and 0.814 (fine-tuned) at stage 1, 0.871 and 0.915 at stage 3, and 0.890 and 0.921 at stage 4: fine-tuning mostly reshapes the last two stages.

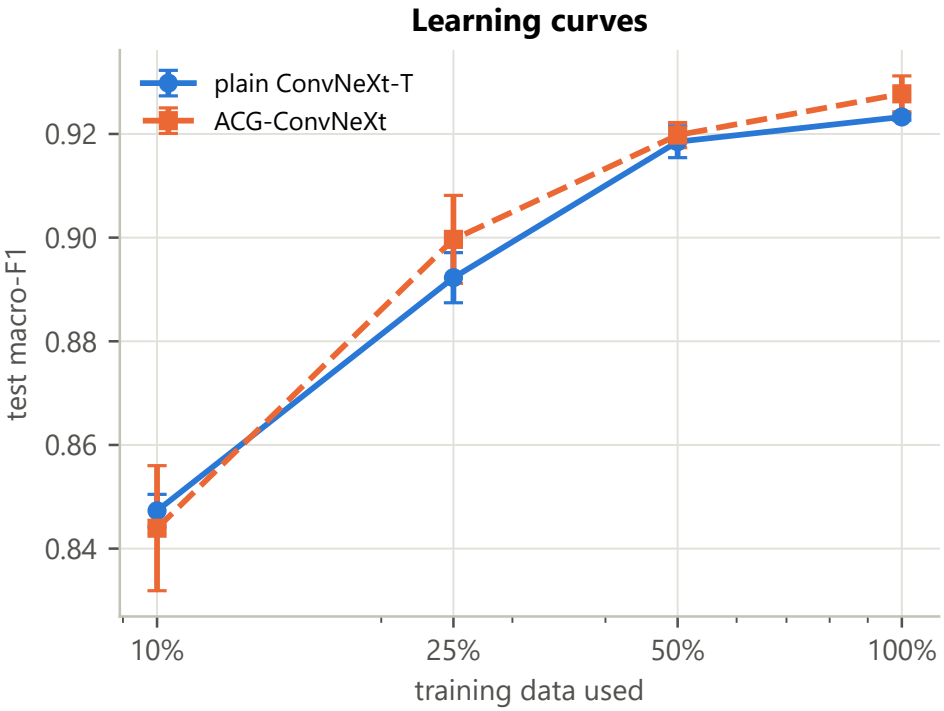


Figure 3.6: Learning curves (mean ± SD over 3 seeds).

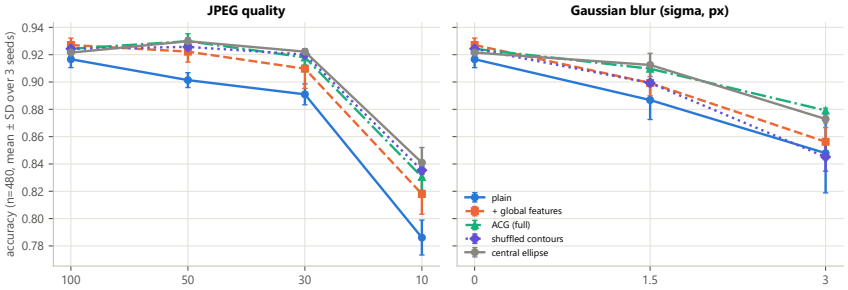


Figure 3.7: Accuracy under JPEG compression and Gaussian blur (mean ± SD over 3 seeds, 480 test frames; contours and features recomputed on the degraded pixels).

3.5 Discussion

Why did the contour not help? Three observations point the same way. (i) The contour is a weak localiser (Dice 0.406, over-segmenting 2.4×), so the spatial prior it provides is only slightly better than a central ellipse. (ii) A fine-tuned ConvNeXt, Swin or DINOv2 backbone already localises the evidence it needs; the Grad-CAM peaks of the plain and guided models fall inside the expert polyp mask equally often (Section V-H). (iii) The residual error is dominated by label ambiguity rather than by missing localisation: 73 % of errors lie in two confusable pairs and 53 % involve flagged labels, so there is little headroom for any prior. The small positive differences we see (≤ 0.6 points) track *extra hand-crafted inputs*, whether or not they involve a contour, and the shuffled and ellipse controls reproduce them. In the classical setting the same pattern holds: contour features beat global features (0.751 to 0.795), but a central ellipse does better still (0.800).

What the study does establish. (1) Kvasir v2 contains duplicates, a label contradiction, strong acquisition shortcuts and substantial label ambiguity, all quantified here. (2) Under one

audited protocol, modern fine-tuned backbones reach macro-F1 of 0.92–0.93 (ensemble 0.9385); published Kvasir accuracies are not directly comparable because class subsets, splits and preprocessing differ and we could not verify those protocols, so we make no claim about ranking against them. Our own control shows a random split changes macro-F1 by about one point (0.9356 against 0.9233, different test images) and the polyp AUROC far more. (3) Calibration is easy to get wrong in practice: validation data reused for model selection under-covered conformal sets (92.9 % against 95 %), and a group-aware split of the test set repaired it. (4) A random split can mislead through the AUROC (0.9994 against 0.9931) more than through accuracy.

Threats to validity and limitations. *Single centre, no patient IDs:* scene groups reduce but cannot exclude patient-level leakage, and nothing here tests transfer to other scopes or hospitals. *Three seeds:* bootstrap intervals over test images capture sampling uncertainty given those seeds; seed-level tests are fragile and are not used for inference. *One recipe for all variants:* deep hyperparameters were not tuned per variant, which is fair across variants but may under-serve any of them (a learning-rate sweep is reported in the repository). *Unsupervised contour only:* a learned, interactive or video-propagated prior could behave differently; we make no claim about those. *Overlay text* inside the field of view was not removed. *Not clinical:* no prospective validation, no regulatory claim, and the dataset terms exclude commercial use without written permission.

Future work. Learned lesion priors trained on Kvasir-SEG, patient-level splits on datasets that provide identifiers, external validation, and video-based consistency are the natural next steps.

3.6 Conclusion

An unsupervised active-contour prior, evaluated with controls, did not measurably improve fine-tuned deep classification of Kvasir findings, while the audit, the leakage-aware protocol, the calibrated and conformal outputs and the error analysis are reusable. We report the null result as a result, with the code, splits and statistics needed to challenge it.

Chapter 4

Engineering, Testing and Deployment

4.1 Repository architecture

- `src/acpolyp/data`: download, ingestion, canonicalisation, curation, dataset and GPU augmentation.
- `src/acpolyp/ac`: salience, two-stage morphological active contour, 136 features, tuning against Kvasir-SEG, extraction.
- `src/acpolyp/models`: ACGNet and ablation modes, backbone factory, frozen-embedding extraction.
- `src/acpolyp/training`: engine (AMP, warm-up/cosine, clipping, early stopping), metrics, calibration, conformal, classical baselines.
- `src/acpolyp/evaluation`: paired statistics, aggregation across seeds, Grad-CAM, figure style.
- `src/acpolyp/app`: FastAPI service and single-page frontend; `inference.py` shares the training preprocessing.
- `scripts/`: experiment table, analysis, figure and paper-asset generation, model bundling, robustness, interpretability, the master script.
- `tests/`: 48 automated tests.

4.2 Engineering decisions driven by evidence

GPU augmentation. DataLoader workers are spawned (not forked) on Windows and each costs about 2.4 GB of commit memory. With train, validation and test loaders persistent, 18 workers exhausted the commit limit (Windows error 1455) and a run hung. Augmentation now runs on the GPU as one affine warp shared by image, contour mask and valid mask, which is faster, removes worker processes and gives the web server the identical evaluation transform.

Lazy memory-mapped caches. Pickling a 1.5 GB array into spawned workers failed; datasets now pickle as file paths and open memory maps on first access.

Unit tests as a safety net. They caught (i) a wrong DeLong AUC (midranks sliced in the wrong order), (ii) YAML 1.1 reading `1e-3` as a string, (iii) a misleading gradient-flow check, because timm zero-initialises the last batch-norm gain of each residual block, so those branches legitimately have zero gradient at step 0, and (iv) a duplicate-ID defect in the dataset itself via the data contract.

Train/serve parity. The bundle is re-run on raw test JPEGs through the complete preprocessing path and compared image by image with the logits saved at training time: 100 % argmax agreement, mean absolute probability difference 2.5×10^{-4} (bf16 versus fp32 arithmetic).

4.3 Reproducibility

Every run directory stores its configuration, git commit, command line, environment, per-epoch history, logits and summary. Seeds cover Python, NumPy, PyTorch and CUDA; cuDNN is deterministic. `torch.use_deterministic_algorithms` is enabled with `warn_only`, so a few CUDA kernels remain non-deterministic: bit-exact replay is not guaranteed, which is why seed-to-seed spread is reported everywhere. `requirements.txt` freezes the environment and `scripts/run_experiments.ps1` rebuilds everything from an empty checkout.

4.4 Web application

A FastAPI backend (`/api/predict`, `/api/results`, `/api/health`, `/api/samples`) serves a dependency-free single-page frontend with a live demo (calibrated probabilities, polyp probability against the validation-tuned threshold, conformal prediction set, contour and Grad-CAM overlays), a pipeline explainer, a results dashboard rendered from the exported JSON, the data audit and an honesty section. Uploads are processed in memory and never stored. The page states prominently that it is a research prototype.

4.5 Packaging

`scripts/build_bundle.py` writes `outputs/artifacts/acpolyp-v1/`: weights (`safetensors`), configuration, scaler, calibration (temperature, polyp threshold, conformal quantile), a model card and SHA-256 hashes. Selection of the bundled model uses *validation* macro-F1 only.

Appendix A

Compliance with the 85-step workflow

#	Step	Status	Where / note
Phase I: Problem formulation, ML framing, feasibility			
1	Domain discovery; why rules fail	done	`docs/FORMULATION.md` §1 (heuristic baseline measured)
2	Task framing, f_θ: X→Y	done	§2
3	Objectives, RQs, H0/H1	done	§3 (RQ1–RQ5)
4	Metric taxonomy (+ cost matrix)	done	§4; `scripts/analyze.py` (5:1 cost operating point)
5	Data/compute/latency feasibility	done	§5
6	Scope, non-goals, MVM vs stretch	done	§6
Phase II: Literature, SOTA, baselines			
7	Literature survey	done	`docs/LITERATURE.md`, `paper/references.bib` (verified entries)
8	Codebase / model-zoo audit	done, limitation	§3 of LITERATURE; no official Kvasir baselines exist to re-run (optional item n/a)
9	Benchmark & leaderboard audit	done, limitation	§1, §4 of LITERATURE; no verifiable public leaderboard for this protocol
10	Gap synthesis & contributions	done	§5 of LITERATURE; paper §I
Phase III: Governance, licensing, planning			
11	Work breakdown & roadmap	done	this file, `scripts/run_deep.py` (declarative experiment table), `scripts/run_experiments.ps1`
12	Risk register & fallback	done	`docs/FORMULATION.md` §7
13	PII sanitisation	n/a (conditional (public images, no user text))	EXIF/comment scan: 0/8000 hits (`data/ingest.py`); burned-in text discussed in `docs/DATA_AUDIT.md`
14	Licence audit	done	`docs/LICENSING.md`

#	Step	Status	Where / note
Phase IV: Environment			
15	Repo structure, git, ignores	done	standard layout; `.gitignore`; `ruff` configured in `pyproject.toml`; `.pre-commit-config.yaml`
16	Locked environment	done	`.venv`, `requirements.txt` (frozen), `pyproject.toml`
17	GPU/OS configuration	done	CUDA torch 2.11+cu128 verified; Windows-specific fixes documented (spawn, commit limit)
18	Run logging	done	`runlog.py`: config, git hash, command, environment, CSV/JSON per run. TensorBoard/MLflow optional n/a
Phase V: Acquisition			
19	Verified idempotent download	done	`data/download.py` (byte count, SHA-256, zip test, zip-slip guard, resume)
20	Ingestion into canonical schema	done	`data/ingest.py` (images → uint8 caches + parquet; no CSV/JSONL inputs exist)
Phase VI: Ground truth			
21	Labelling schema	n/a (conditional (dataset labels used as given))	class taxonomy in `data/contracts.py`; boundary cases (esophagitis↔z-line) in the audit
22	Self-annotated gold set	n/a (optional (no clinical expertise to annotate honestly))	Kvasir-SEG expert masks serve as the gold set for the contour stage
23	Programmatic labelling	n/a (optional)	cleanlab used for *flagging*, not relabelling
Phase VII: Cleaning and QA			
24	Sanitisation	done	decode check, valid-FOV, artefact masking (`data/preprocess.py`)
25	Duplicates	done	SHA-256 + embedding cosine (pHash evaluated and rejected, with evidence)
26	Outliers / label noise	done	range checks, IQR blur outliers, `cleanlab` (550 flagged)
27	Schema validation	done	`data/contracts.py` (pydantic + assertions); it caught the duplicate-ID defect
Phase VIII: EDA			
28	Distributions, normality	done	`scripts/eda.py`; `docs/DATA_AUDIT.md` §4
29	Imbalance, subgroups	done	1.007 (8-class), 1:7 (polyp-vs-rest), per-format and per-box subgroups
30	Collinearity, shortcuts	done	VIF, Spearman, MI; resolution/box probes
31	Manifold	done	UMAP (`outputs/figures/eda_umap.*`)
Phase IX: Splits and leakage			

#	Step	Status	Where / note
32	Deterministic split	done	stratified group split, seed 42; random split as control; OOD hold-out optional n/a
33	Leakage audit	done	`curate.leakage_audit`; zero group/hash overlap asserted
34	CV folds	done	5 grouped folds serialised (`cv_fold`), used for out-of-fold label-noise estimation
35	Versioned artefacts	done	`data/processed/train,val,test_ids.txt` + SHA-256 in `curation_report.json`; Arrow/HDF5 optional n/a (memory-mapped `.npy` used)
Phase X: Features and preprocessing			
36	Scaling	done	train-only 1/99 winsorise + z-score (`dataset.fit_scaler`); power transforms optional n/a
37	Categorical/temporal encoding	n/a	no such features
38	Domain features, selection	done, limitation	136 features (colour, LBP, GLCM, shape, boundary); importance by MI and SHAP rather than pruning
39	Tokenisation	n/a	vision task
40	Augmentation	done	GPU affine + colour jitter, contour mask follows geometry (`data/augment.py`, tested)
41	Imbalance mitigation	n/a	classes balanced; polyp-vs-rest handled by threshold tuning, not resampling
42	Pipeline serialisation	done	`scaler.json`, `calibration.json`, `config.yaml` inside each bundle
Phase XI: Modelling			
43–46	Formulation, loss, architecture, parameters/VRAM	done	`docs/FORMULATION.md` §8; `models/acgnet.py`; measured params and VRAM in profiling table
Phase XII: Skeleton and baselines			
47	Dataset/DataLoader	done	`data/dataset.py` (uint8 contract, in-process loader; spawn workers rejected on Windows)
48	Walking skeleton	done	`tests/test_walking_skeleton.py` (constant model, real loader, runs in seconds)
49	Trivial & heuristic baselines	done	random, prior, Lab-centroid heuristic (`classical.run_trivial`)
50	Classical baselines	done	LR, RBF-SVM, RF, LightGBM, XGBoost
51	Pretrained / confidence baselines	done	frozen-DINOv2 linear probe; max-softmax risk–coverage
Phase XIII: Implementation			
52	Config schema	done	`configs/*.yaml` + dotted CLI overrides (unit-tested; caught the YAML `1e-3` string bug)
53	Modular architecture, hooks	done	`ACGNet`; forward hooks for Grad-CAM and layer-wise probes

#	Step	Status	Where / note
54	Metrics	done	`training/metrics.py` (`ignore_index` n/a no padding)
55	Optimiser, scheduler, clipping	done	AdamW, no decay on bias/norm, warm-up + cosine, clip 1.0
56	Train/eval loops	done	`training/engine.py` (accumulation optional n/a not needed)
57	Seeding	done, limitation	all RNGs seeded; cuDNN deterministic; `use_deterministic_algorithms(warn_only=True)`, so a few CUDA kernels remain non-deterministic (bit-exact replay is not guaranteed; seed-to-seed spread is reported)
58	Checkpoint, early stopping	done	best `.safetensors` by val macro-F1, patience 4
Phase XIV: Debugging			
59	Shape/device asserts	done	in `ACGNet.forward`, `predict` (finite logits) and tests
60	Single-batch overfit	done	`tests/test_model_sanity.py`
61	Gradient flow	done	test (after one step, because timm zero-inits last BN gains); grad norm logged each epoch
62	Numerical stability	done	fp32 loss on bf16 logits, finite asserts, stable softmax
63	Unit tests	done	48 tests; batch-order invariance included
Phase XV: Optimisation			
64	AMP, SDPA, cached features	done	bf16 autocast; timm fused attention for ViT/Swin; cached frozen embeddings
65	PEFT	limitation (optional)	frozen-backbone linear probe only; LoRA not used
66	HPO	done, limitation	Optuna/TPE on validation for the contour (80 trials) and classical models (25 each); deep nets use one shared recipe. A learning-rate sweep (seed 0) and a post hoc 3-seed replication at 2x LR quantify what that costs: the plain model improves by 0.8 points at 2x LR and the guided-vs-plain gap stays null
67	Ensembling	done	seed and backbone soft-voting
Phase XVI: Post-processing			
68	Threshold tuning	done	Youden J on validation (+ cost-optimal threshold)
69	Calibration	done	temperature scaling, reliability diagrams, ECE, Brier
70	Uncertainty	done	split-conformal (LAC), risk–coverage
Phase XVII: Evaluation			
71	Locked test evaluation	done	test touched once per frozen run; secondary dataset optional n/a (HyperKvasir overlaps)
72	Multi-seed	done	deep N=3, classical N=5, mean ± SD
73	Significance	done	paired bootstrap (2,000), McNemar, DeLong, Holm, Cohen's d, seed-level tests

#	Step	Status	Where / note
74	Ablations	done	7 variants + 3 controls
75	Sensitivity, scaling, Pareto	done	learning curves, LR sweep, accuracy–latency
76	OOD / perturbation	done	JPEG compression and Gaussian blur on 480 stratified frames, 3 seeds x 5 variants incl. controls (`scripts/robustness2.py`); single-seed noise/blur/brightness/JPEG screen (`scripts/robustness.py`, helper)
77	Profiling	done	latency (warm-up + sync), VRAM, train time, checkpoint size
Phase XVIII: Interpretability and errors			
78	Confusion, per-class, ROC/PR	done	`analyze.py`
79	Failure taxonomy	done	error gallery + automated categories (label-flagged, confusable pairs) + manual inspection notes in the report
80	Attribution	done	TreeSHAP, Grad-CAM (pointing game vs expert masks)
81	Layer-wise probes	done	per-stage linear probes, pretrained vs fine-tuned
82	Subgroup audit	done	by scope format, ScopeGuide box, class; worst-group accuracy
Phase XIX: Packaging			
83	One-command CLI	done	`scripts/run_experiments.ps1`; entry points `scripts/train.py`, `scripts/evaluate.py`, `scripts/predict.py` (wrappers over `acpolyp.training.train` / `acpolyp.cli`), `python -m acpolyp.cli serve`
84	Bundle + local UI	done	`outputs/artifacts/acpolyp-v1/` (weights, scaler, calibration, model card, hashes); FastAPI + custom single-page UI (instead of Gradio)
Phase XX: Documentation and archival			
85	README, figures/tables script, manuscript, slides, clean-room check	done, limitation	`README.md`; `scripts/make_figures.py` (runs EDA, contour, analysis, export, paper-asset scripts: 300-dpi vector PDFs, `booktabs` tables with bold-best and ± SD); `paper/main.tex` (IEEEtran, 7 pp.); `report/report.pdf` (34 pp.); `slides/ACPolyp.pptx` (13 slides); clean-room clone + install + 48 tests run locally (the 8 h pipeline was not re-run from scratch). Publishing to GitHub/Zenodo/Hugging Face is left to the owner (needs their accounts)

Appendix B

Failure Taxonomy of the Most Confident Errors

The 36 most confident errors of the seed-ensembled ACG-ConvNeXt (117 errors in total on the test split) were inspected and sorted into four categories. *The categories are visual judgements by the project author, who is not a clinician; they are not a clinical adjudication and should be reviewed by a gastroenterologist.*

Category	n	flagged	What the frames show
A. Esophagitis ↔ z-line	17	15	Irregular, inflamed or whitish mucosa at or near the gastro-oesophageal junction; both labels plausible, esophagitis is defined relative to the z-line.
B. Dyed lifted ↔ resection margin	7	5	Blue-dye craters or lifted lesions at different stages of endoscopic resection; the visible state often supports either label.
C. "Polyp" predicted as another class	7	2	Frames labelled polyps with no distinct visible lesion (homogeneous mucosa) or a small or marginal lesion; predicted cecum, colitis or pylorus. The clinically important group.
D. Other	5	4	Diffuse redness labelled colitis predicted as cecum or esophagitis; a snare loop with bleeding.

Table B.1: Failure taxonomy of the 36 most confident errors; "flagged" = flagged by confident learning. Overall 26 of 36 are flagged, against about 5 % of correct predictions.

Appendix C

Additional Figures

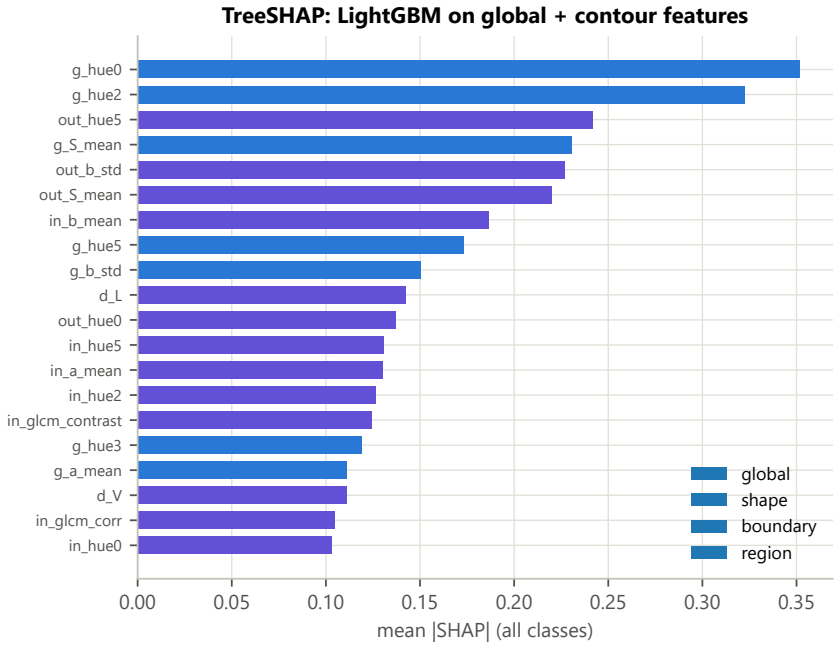


Figure C.1: TreeSHAP importance of the 20 most influential features of the LightGBM model on global and contour features (all classes). Region (inside/outside colour and texture) and global colour features dominate; contour shape and boundary features contribute about 6 % and 3 % of the total.

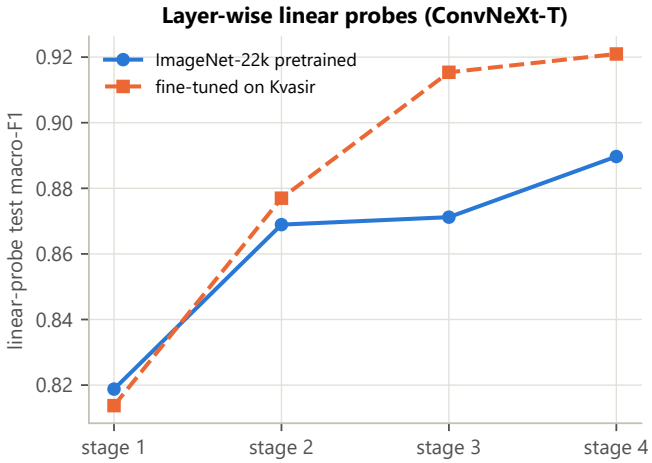


Figure C.2: Layer-wise linear probes on ConvNeXt-T stages: ImageNet-22k pretrained versus fine-tuned on Kvasir. Fine-tuning mostly reshapes the last two stages.

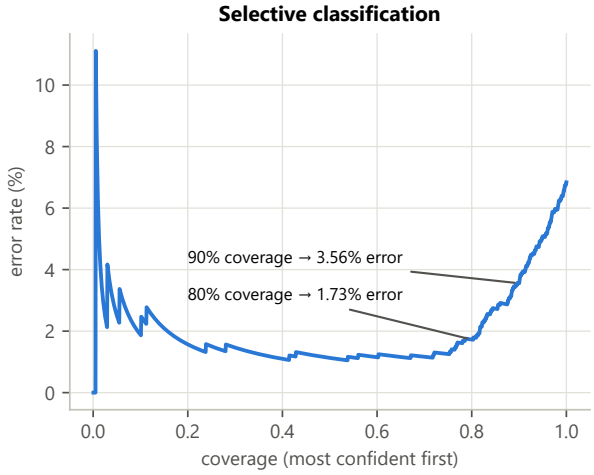


Figure C.3: Selective classification: error rate on the most confident fraction of test frames (ACG-ConvNeXt, seed 0).

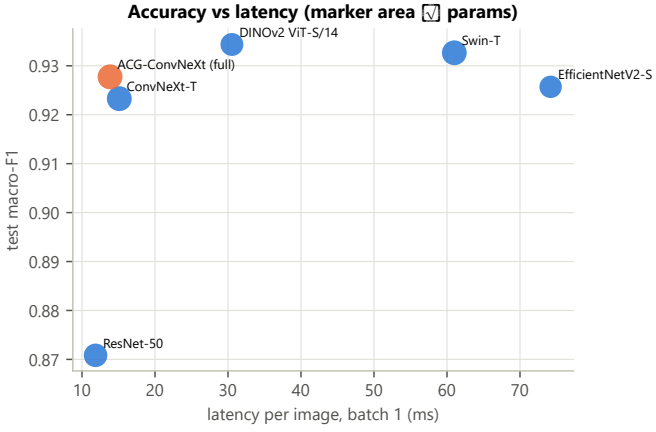


Figure C.4: Accuracy–latency trade-off (batch-1 latency on an RTX 4060 Laptop GPU; marker area proportional to parameter count).

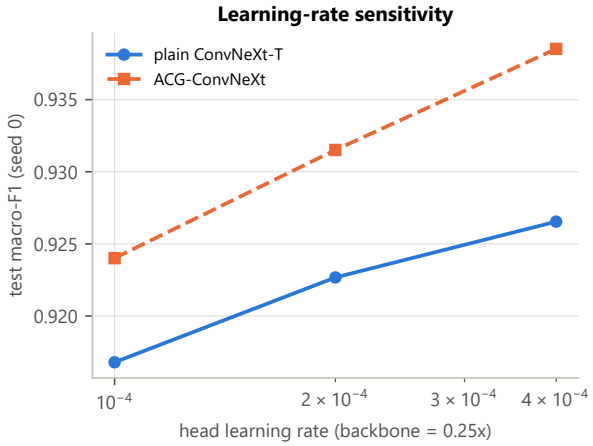


Figure C.5: Learning-rate sensitivity for plain and guided ConvNeXt-T (seed 0). A three-seed replication at twice the default rate is reported in Section 3.2.1 of the results text.

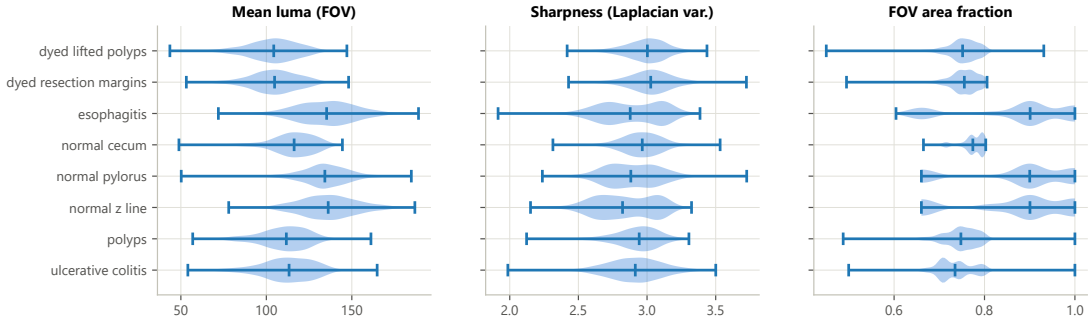


Figure C.6: Image statistics by class: mean luma, sharpness and field-of-view fraction.

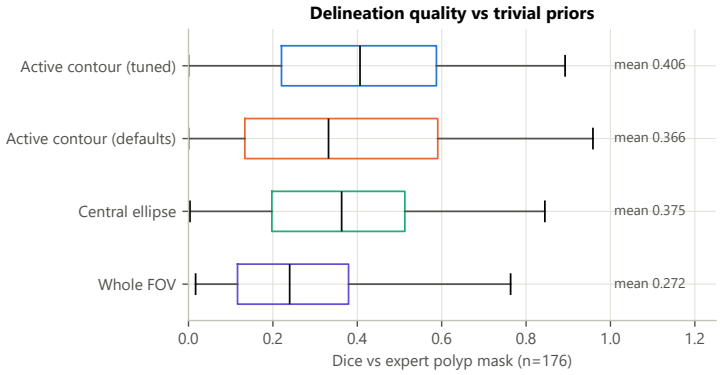


Figure C.7: Delineation quality: tuned contour versus trivial priors on 176 expert-masked test-split polyps.

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